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C1C1 Pathway DesignerMulti-Substrate C1 Pathway Platform

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About

C1 Pathway Design

Designed for CO₂, methanol, formaldehyde, and formate substrates, the platform systematically screens C1 assimilation pathways to target products through candidate-reaction combination enumeration and pRBA feasibility analysis.

Start Calculation

View Results

Workflow

01

Select Substrate and Product

Choose a C1 substrate and the target molecule to produce.

02

Set Combination Size

Specify the number of candidate reactions opened simultaneously.

03

Enumerate Combinations

Generate combinations from the substrate-specific candidate pool.

04

pRBA Feasibility Screening

Determine whether each combination supports target production.

05

Analyze and Visualize

Output fluxes, thermodynamic plots, MDF plots, and pathway networks.

Platform Capabilities

Unified Multi-Substrate Framework

Supports CO₂, methanol, formaldehyde, and formate as C1 substrates.

Reaction Combination Search

Enumerates substrate-specific candidate pools and compares pathway feasibility.

Flux and Thermodynamics

Provides pRBA flux tables, standard reaction Gibbs energy plots, and MDF plots.

Visualization and Annotation

Displays pathway networks with reaction equations, fluxes, and EC numbers.

C Calculation results

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Calculation Results

Browse and manage calculation tasks, expand details to inspect candidate pathways, and open pathway visualizations.

All 69

Running 0

Completed 69

Search by job ID, substrate, or product

Refresh

job id	substrate	product	C1 reactions per combination	maximum pathway reactions	status	timestamp	action
efb66b5bde	Formaldehyde	3-Phospho-D-glycerate (C00197)	1	20	Completed	2026-07-27T10:16:06	detail delete

Pathway Results (3)

Collaps Details

Pathway ID	Substrate	Product	Reaction Count	Delta G	MDF	action
path_0001	Formaldehyde	3-Phospho-D-glycerate (C00197)	6	-102.76 kJ/mol	19.19 kJ/mol	Visualize Download
path_0002	Formaldehyde	3-Phospho-D-glycerate (C00197)	12	-42.98 kJ/mol	6.00 kJ/mol	Visualize Download
path_0003	Formaldehyde	3-Phospho-D-glycerate (C00197)	12	-43.79 kJ/mol	7.00 kJ/mol	Visualize Download

efb66b5b7e2	Formaldehyde	3-Phospho-D-glycerate (C00197)	1	20	Completed	2026-07-27T20:54:42	detail delete
1bf118529f2d6	Formaldehyde	3-Phospho-D-glycerate (C00197)	1	20	Completed	2026-07-28T10:07:01	detail delete
62b5db77a8fc	Formaldehyde	3-Phospho-D-glycerate (C00197)	1	20	Completed	2026-07-28T10:04:18	detail delete

E Pathway network

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Pathway Network

Show Calculations

NO

Download Pathway Table

Save JSON

Show Reaction Nodes

Download SVG

PEP

3.33

PYR

3.33

DHA

3.33

DHAP

3.33

GAP

3.33

13DPG

3.33

G3P

B Task submission

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Submit a Pathway Search Task

Select a C1 substrate, target product, candidate-reaction combination size, and maximum pathway length. The system enumerates the complete substrate-specific candidate pool and uses pRBA to identify all qualifying feasible pathways.

Task Parameters

Substrate

Formaldehyde

Product

3-Phospho-D-glycerate (C00197)

C1 Reactions per Combination

1

Maximum Pathway Reactions

20

Email Notification

Optional, e.g. user@example.com

Submit Task

Reset Parameters

Parameter Guide

Substrate / Product

Four C1 substrates are currently supported.

C1 Reactions per Combination

Simultaneously opens n candidate reactions and tests whether the combination supports production of the target product.

Maximum Pathway Reactions

Limits all feasible pathways whose length is less than or equal to this main-pathway length.

Email Notification

If SMTP is configured, a notification is sent when the task finishes; otherwise, the address is only recorded.

D Thermodynamics and MDF

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Pathway Visualization

Select a pathway on the Results page to view thermodynamic plots, the MDF plot, the flux network, and reaction details.

Pathway ID

path_0001

Substrate

Formaldehyde

Product

3-Phospho-D-glycerate (C00197)

Reactions

6

Delta G

-102.76 kJ/mol

MDF

19.19 kJ/mol

Thermodynamics and MDF

Standard Gibbs Energy

Download

MDF

Download

Reaction step

Reaction Step

F Reaction details

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Reaction Details

6 rows

Reaction ID	Equation
FLS	3,0 FORMALDEHYDE_c -> DIHYDROACETONE_c
PHOSPHOGlycerate-KINASE-GTP-RIN	GTP_c + GTP_c <=> GMP_c + GDP_c
CAPOINOSHYCHIN-RIN	GAP_c + NAD_c > R_c <=> DAP_c + NADH_c + PROTON_c
RIN-14117	GTP_c + PYRUVATE_c -> GDP_c + PHOSPHOENOLPYRUVATE_c + PROTON_c
TRIOPHOSPHORIZATION-RIN	GAP_c <=> DIHYDROXYACETONE_c + PHOSPHATE_c
2,7,12,12-RIN	DIHYDROACETONE_c + PHOSPHOENOLPYRUVATE_c -> DIHYDROXYACETONE_c + PHOSPHATE_c + PYRUVATE_c

Reaction Details

Reaction ID

PHOSPHOGlycerate-KINASE-GTP-RIN

EC number

2.7.2.10

Plus

-3.33333

Standard AG

-22.018 kJ/mol

Reaction

GDP_c + GTP_c <=> GMP_c + GDP_c